

Homework 7

Oct 29, 2024

1 (50'). Prove that the class of languages in P is closed under the following operations:

- (a) union
- (b) intersection
- (c) concatenation

2 (50'). Let

$L = \{(\alpha, x) : \text{On input } x, \text{ TM } M_\alpha \text{ uses at most } n^2 \text{ space, and loops forever}\}.$

Prove L is decidable.